

Modeling Bottom-up Information Quality during Language Processing

自下而上信息质量建模在语言处理中的应用

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Abstract

摘要

Contemporary theories model language processing as integrating both top-down expectations and bottom-up inputs. One major prediction of such models is that the quality of the bottom-up inputs modulates ease of processing—noisy inputs should lead to difficult and effortful comprehension. We test this prediction in the domain of reading. First, we propose an information-theoretic operationalization for the “quality” of bottom-up information as the mutual information (MI) between visual information and word identity. We formalize this prediction in a mathematical model of reading as a Bayesian update. Second, we test our operationalization by comparing participants’ reading times in conditions where words’ information quality has been reduced, either by occluding their top or bottom half, with full words. We collect data in English and Chinese. We then use multimodal language models to estimate the mutual information between visual inputs and words. We use these data to estimate the specific effect of reduced information quality on reading times. Finally, we compare how information is distributed across visual forms. In English and Chinese, the upper half contains more information about word identity than the lower half. However, the asymmetry is more pronounced in English, a pattern which is reflected in the reading times.

现代理论将语言处理建模为整合自上而下的预期和自下而上的输入。这类模型的一个主要预测是：自下而上的输入质量会调节处理的难易程度——质量较差的输入会导致理解 and 处理更加困难和费力。我们在阅读领域检验了这一预测。首先，我们提出了一种信息论的操作化方法，将“自下而上信息的质量”定义为视觉信息与词身份之间的互信息（MI）。我们将这一预测形式化为一个阅读的数学模型，即贝叶斯更新。其次，我们通过比较在词的信息质量被降低的条件下（通过遮挡词的上半部分或下半部分）参与者阅读时间，来检验我们的操作化方法。我们收集了英语和中文的数据。然后，我们使用多模态语言模型来估计视觉输入与词之间的互信息。我们利用这些数据来估计信息质量降低对阅读时间的具体影响。最后，我们比较了信息在视觉形式中的分布情况。在英语和中文中，上半部分包含比下半部分更多的词身份信息。然而，这种不对称性在英语中更为显著，这种模式在阅读时间中也有所体现。



Figure 1: images/8941e00f3ac66101f8fb266fe6450f1dcc39ff73fb9cc9cdac8df7990b4fc925.jpg

<https://github.com/DiLi-Lab/>

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Bottom-Up-Information.git

Bottom-Up-Information.git

1 Introduction

1 引言

During reading, individuals actively expend cognitive effort to extract information. Many contemporary theories of language comprehension in general, and reading in particular, model this process as a rational integration of bottom-up and top-down information (Legge et al., 1997; Norris, 2006; Bicknell and Levy, 2010; Gibson et al., 2013; Gauthier and Levy, 2023). Bottom-up information refers to the perceptual input (e.g., visual forms

在阅读过程中，人们主动付出认知努力以提取信息。许多关于语言理解的一般理论，特别是关于阅读的理论

论, 将这一过程建模为自下而上和自上而下信息的理性整合 (Legge 等, 1997; Norris, 2006; Bicknell 和 Levy, 2010; Gibson 等, 2013; Gauthier 和 Levy, 2023)。自下而上的信息指的是知觉输入 (例如, 视觉形式)

of words), while top-down information includes the prior beliefs and expectations about what messages or word-forms are likely to be encountered, and is guided by the reader's linguistic and contextual knowledge. A central prediction of such models is that the ease of reading should be influenced by the quality of the bottom-up information. In the modality of visual reading, visual signals that effectively convey information about the intended message are expected to facilitate fast and effortless comprehension (Balota, 1994). Conversely, degraded visual signals—caused by factors such as lighting, occlusion, or visual interference—are likely to increase processing effort and raise the likelihood of errorful reading.

(词语) 的处理, 而自上而下的信息包括读者对即将遇到的信息或词语形式的先验信念和期望, 并受到读者语言和上下文知识的指导。这类模型的一个核心预测是, 阅读的容易程度应受到自下而上信息质量的影响。在视觉阅读模式中, 能够有效传达预期信息的视觉信号预计会促进快速而轻松的理解 (Balota, 1994)。相反, 由于光照、遮挡或视觉干扰等因素导致的视觉信号退化, 很可能会增加处理努力, 并提高错误阅读的可能性。

This prediction fits well within noisy channel models of reading. In a noisy channel model (Shannon, 1948), a message is encoded and sent over a channel, where it is potentially corrupted. A receiver, at the other end of the channel, must decode the most probable intended message given the received inputs. Previous work has looked at the role of noise during reading, demonstrating how noise over uncertain inputs can lead to non-veridical interpretations (Levy, 2008b; Gibson et al., 2013).

这种预测很好地符合了阅读中的噪声通道模型。在噪声通道模型 (Shannon, 1948) 中, 一条信息被编码并通过一个通道发送, 在传输过程中可能会受到干扰。在通道的另一端, 接收者必须根据接收到的输入, 解码出最可能的意图信息。先前的研究探讨了阅读过程中噪声的作用, 展示了噪声如何导致对不确定输入的非真实解释 (Levy, 2008b; Gibson et al., 2013)。

While intuitive, to the best of our knowledge, the impact of noisy inputs on reading effort has not been quantified within a formal computational model of reading. That is, although many theories of reading assume that poorer sensory input leads to more effortful processing, and classic experimental work has shown that reduced visual signals increase processing difficulty and interact with other lexical properties (Rumelhart and Siple, 1974), they have not derived or tested this relationship quantitatively. In this paper, we aim to fill this gap by providing an information-theoretically grounded, quantitative account of how bottom-up input quality affects processing effort. Our central proposal is that input quality can be formalized as the mutual information (MI) between (visual) input

虽然直观, 据我们所知, 噪声输入对阅读努力的影响尚未在阅读的正式计算模型中进行量化。也就是说, 尽管许多阅读理论假设较差的感官输入会导致更费力的处理, 而且经典实验工作已经表明, 减少的视觉信号会增加处理难度, 并与其他词汇特性相互作用 (Rumelhart 和 Siple, 1974), 但它们并未定量地推导或检验这种关系。在本文中, 我们旨在通过提供一个基于信息论的、定量的解释, 来填补这一空白, 即说明自下而上的输入质量如何影响处理努力。我们的核心观点是, 输入质量可以形式化为 (视觉) 输入之间的互信息 (MI)。

and word identity. From an information-theoretic perspective, a signal is informative to the extent that it reduces uncertainty about a target variable—in this case, the identity of a word. We assume that greater processing effort manifests in longer reading times, and therefore predict that reductions in mutual information should lead to systematic slowdowns in reading.

信息论视角下的词身份。从信息论的角度来看, 一个信号的信息量取决于它对目标变量的不确定性减少程度——在这种情况下, 就是词的身份。我们假设更大的处理努力会表现为更长的阅读时间, 因此预测信息互减量的减少会导致阅读速度的系统性减慢。

This paper makes three contributions: First, we instantiate the above operationalization of visual input quality in reading under a formal model of reading as a Bayesian update. Second, we

provide a quantitative estimate of the cost of reduced input quality on processing effort. To do so, we use multimodal language models to estimate mutual information over a dataset of partially masked word images. We then collect human reading times on the same stimuli, using the MoTR paradigm (Wilcox et al., 2024), which simulates eye-tracking, and can be used to collect data over the web. We use these data to estimate the relationship as a specific slowdown in terms of nats of information gain (the pointwise variant of mutual information) per millisecond of processing time. Our data suggest that the cost of reduced information is not linear—small losses in informational quality can lead to disproportionately large increases in reading time, particularly in the upper regions of a signal’s informational range.

本文的贡献有三点：第一，我们在一个形式化的阅读模型（将阅读视为贝叶斯更新）中实例化了上述关于阅读中视觉输入质量的操作化定义。第二，我们提供了输入质量降低对处理努力成本的定量估计。为此，我们使用多模态语言模型来估计部分遮蔽单词图像数据集上的互信息。然后，我们使用 MoTR 框架 (Wilcox 等, 2024) 收集人类的阅读时间，该框架模拟了眼动追踪，可用于在互联网上收集数据。我们利用这些数据来估计关系，即以每毫秒处理时间的信息增益（互信息的点态变体）为单位的具体减缓程度。我们的数据表明，信息减少的成本并非线性——信息质量的微小损失可能导致阅读时间出现不成比例的大幅增加，尤其是在信号信息范围的上部区域。

Our third contribution is to compare how information is distributed across visual forms of words in two typologically distinct languages. To that end, we collect data in both English and Chinese, representing alphabetic and logographic scripts, respectively. We find that, in both languages, the upper half of a word contains more information about word identity than the lower half. However, the asymmetry is more pronounced in English than in Chinese, a pattern that is reflected in the reading times.

我们的第三项贡献是比较两种类型学上不同的语言中，信息在单词的不同视觉形式上的分布情况。为此，我们分别收集了英语和汉语的数据，分别代表了字母文字和意音文字。我们发现，在这两种语言中，单词的上半部分比下半部分包含更多关于单词身份的信息。然而，这种不对称性在英语中比在汉语中更为明显，这种模式在阅读时间上得到了体现。

2 Formal Model

2 形式模型

2.1 Reading as Bayesian Update

2.1 阅读作为贝叶斯更新

Following an extensive prior literature (Norris, 2006; Bicknell and Levy, 2010; Gauthier and Levy, 2023), we model word recognition as a Bayesian update process. Readers incrementally process a word w drawn from a vocabulary W , where $w \in W$ denotes a realization of a random variable W taking values in W . We refer to a word at a particular timestep, t , as w_t and the corresponding random variable at this timestep as W_t . We assume that readers intake individual samples of input e , where $e \in R$ denotes a realization of a random variable E ranging over the samples¹. Input samples could be either a patch of visual input for visual reading or a haptic percept in the case of braille. Following previous work (e.g., Bicknell and Levy, 2010), we model the process of reading as one of sequential word identification given input e and a previous context of words $w_{<t}$. In such models, readers are assumed to rationally integrate their prior expectations about a word, $P(w_t | \mathbf{w}_{<t})$, with the likelihood of the observed input e , $P(e | w_t, \mathbf{w}_{<t})$. Instead of a single sample, we assume that readers integrate evidence over k samples, $e_{1:k}$. The rational update process we use to model reading is therefore:

在广泛的先前文献 (Norris, 2006; Bicknell and Levy, 2010; Gauthier and Levy, 2023) 的基础上，我们将单词识别建模为一个贝叶斯更新过程。读者逐步处理一个从词汇表 W 中抽取的单词 w ，其中 $w \in W$ 表示随机变量 W 在 W 中的一个实现。我们称在特定时间步 t 的一个单词为 w_t ，并称在该时间步对应的

随机变量为 W_t 。我们假设读者接收输入样本 e 的个体实例，其中 $e \in R$ 表示随机变量 E 在样本集合¹ 上的一个实现。输入样本可以是视觉阅读中的一个视觉输入片段，或者是盲文中的触觉感知。根据先前的研究（例如，Bicknell and Levy, 2010），我们将阅读过程建模为在输入 e 和先前单词上下文 $w_{<t}$ 的基础上进行的连续单词识别过程。在这些模型中，假设读者理性地将他们对单词的先验期望 $P(w_t | \mathbf{w}_{<t})$ 与观察到的输入 e 的似然 $P(e | w_t, \mathbf{w}_{<t})$ 相结合。而不是单一的样本，我们假设读者在 k 个样本上整合证据，即 $e_{1:k}$ 。因此，我们用来建模阅读的理性更新过程是：

$$P(w_t | \mathbf{e}_{1:k}, \mathbf{w}_{<t}) \propto \quad (1)$$

$$P(w_t | \mathbf{w}_{<t}) \times \prod_{i=1}^k P(e_i | w_t, \mathbf{w}_{<t})$$

This tells us how readers update beliefs about a word given inputs and priors. But reading is a dynamic process. How do readers choose when to move on to the next word? Previous work models this by proposing that readers draw samples until the uncertainty about the current word reaches a threshold, ϕ , at which point they move on (e.g., Li and Futrell, 2024). We quantify uncertainty as the entropy of the posterior distribution. That is, sampling continues until:

这告诉我们读者如何根据输入和先验知识来更新对某个词的信念。但阅读是一个动态的过程。读者如何决定何时继续到下一个词？先前的研究通过提出读者持续抽样，直到当前词的不确定性达到一个阈值 ϕ ，此时他们便继续到下一个词（例如，Li 和 Futrell, 2024）。我们将不确定性量化为后验分布的熵。也就是说，抽样会持续直到：

$$H(P(w_t | \mathbf{e}_{1:k}, \mathbf{w}_{<t})) \leq \phi \quad (2)$$

However, given a particular actual input w^* we cannot be certain how many samples a reader draws or what information each sample contains. To account for this uncertainty, we therefore make the prediction that readers will move on when the expected entropy falls below this threshold, where the expectation is taken over uncertain inputs:

然而，给定一个特定的实际输入 w^* ，我们无法确定读者会抽取多少个样本，或者每个样本包含哪些信息。为了应对这种不确定性，因此我们做出预测：当预期的熵值低于这一阈值时，读者会继续阅读，其中期望值是针对不确定输入的：

$$\mathbb{E}_{\mathbf{E}_{1:k}} [H(W_t | \mathbf{E}_{1:k}, \mathbf{w}_{<t})] \leq \phi \quad (3)$$

Although we assume that reading does take place given a context, for the rest of this section, we will drop the word-context term, $w_{<t}$. We note that it would be easy to add this term back into the subsequent equations as a conditioning variable without changing the overall model.

虽然我们假设在给定上下文的情况下阅读是会发生，但在本节的其余部分，我们将省略“词-上下文”这一术语，即 $w_{<t}$ 。我们指出，很容易在随后的方程中将这一术语作为条件变量重新加入，而不会改变整体模型。

2.2 Quality of Bottom-Up Evidence

2.2 自下而上证据的质量

We model the quality of the inputs as the mutual information between the inputs and the word identity, i.e., as $I(W; E)$. High-quality inputs do a better job of reducing uncertainty over words.

For a given word-identification step, we can write the mutual information between a word and the total number of samples drawn as $I(W; \mathbf{E}_{1:k})$. Using the chain rule of mutual information (Cover, 1999) and assuming that there is conditional independence between samples, given W , we can derive the following inequality:²

我们将输入质量建模为输入与单词身份之间的互信息，即 $I(W; E)$ 。高质量的输入在减少单词不确定性方面表现更好。对于给定的单词识别步骤，我们可以将单词与总样本数之间的互信息表示为 $I(W; \mathbf{E}_{1:k})$ 。利用互信息的链式法则 (Cover, 1999)，并假设在给定 W 的条件下样本之间条件独立，我们可以推导出以下不等式：²

$$I(W; \mathbf{E}_{1:k}) = \sum_{i=1}^k I(W; E_i | \mathbf{E}_{1:i-1}) \quad (4a)$$

$$\text{assuming} \quad \leq \sum_{i=1}^k I(W; E_i) \quad (4b)$$

$$\text{cond. independence} \quad \leq k \times I(W; E); \quad (4c)$$

How is the mutual information between inputs and words related to the reading process, as described above? We assume that taking samples and processing these samples takes cognitive effort. Following previous work, we also assume a link between effort and time (Levy, 2008a; Hale, 2001). Therefore, the more samples, k , a reader needs to take in order to reduce uncertainty, the longer it will take them to read a given word.

如上所述，输入和词语之间的互信息与阅读过程有何关系？我们假设采取样本并处理这些样本需要认知努力。根据以往的研究，我们还假设努力与时间之间存在联系 (Levy, 2008a; Hale, 2001)。因此，读者需要采取更多样本 k 来减少不确定性，那么他们阅读某个词语所需的时间就会更长。

We can now link the quality of inputs to our reading process through the definition of mutual information:

我们现在可以通过互信息的定义，将输入的质量与我们的阅读过程联系起来：

$$I(W; \mathbf{E}_{1:k}) = H(W) - H(W | \mathbf{E}_{1:k}) \quad (5)$$

Plugging in the inequality from 4c, and the definition of conditional entropy,³ we rearrange the terms:

将 4c 中的不等式以及条件熵的定义代入，³ 我们重新排列各项：

$$\mathbb{E}_{\mathbf{E}_{1:k}} [H(W | \mathbf{E}_{1:k})] \geq H(W) - k \times I(W; E) \quad (6)$$

That is, the expected entropy of the posterior distribution, given uncertain inputs, is greater than the entropy over words minus the number of samples taken times the mutual information between the samples and the words.

也就是说，在给定不确定输入的情况下，后验分布的预期熵大于词的熵减去所取样本数乘以样本与词之间的互信息。

For our model of reading, we are interested in when the entropy of the posterior distribution is approximately ϕ . In particular, we are interested

对于我们的阅读模型，我们关注的是后验分布的熵何时大约为 ϕ 。特别是，我们关注

in how many samples must be drawn to reach this threshold, as this determines the effort (and therefore the time) required to reduce uncertainty enough to move on to the subsequent word. Substituting in our threshold parameter in and rearranging the terms, we have:

为了达到这个阈值，必须抽取多少个样本？因为这决定了减少不确定性所需的努力（因此是时间），从而继续处理下一个单词。将我们的阈值参数代入并重新排列各项，我们得到：

$$k \geq \frac{H(W) - \phi}{I(W; E)} \quad (7)$$

The minimum number of samples required to reach the threshold grows with the entropy of the distribution over W . Likewise, it decreases with the mutual information between W and E . Because we assume a link among the number of samples, effort and time, this leads us to the following two predictions:

达到阈值所需的最小样本数随着 W 上分布的熵而增长。同样，它也跟着 W 和 E 之间的互信息而减少。由于我们假设样本数量、努力和时间之间存在联系，这导致我们得出以下两个预测：

Prediction 1 Top-Down Processing & Entropy: As the entropy of a word-position W increases, average reading time increases.

预测 1 自上而下处理与熵：随着词位 W 的熵增加，平均阅读时间增加。

Prediction 2 Bottom-up Processing & Mutual Information: As the mutual information between words W and their visual representations E decreases, average reading time increases.

预测 2 自底向上处理与互信息：随着词语 W 与其视觉表示 E 之间的互信息减少，平均阅读时间增加。

In fact, Prediction 1 has already been investigated by Pimentel et al. (2023), whose results confirm our prediction. Pimentel et al. refer to the entropy over the next word, given a set of previous words $H(W_t | \mathbf{w}_{<t})$ as a word’s contextual entropy. They find that as word-level contextual entropy increases, so too does reading time. For the rest of this paper, therefore, we are interested in testing Prediction 2, namely whether the quality of bottom-up evidence, modeled as mutual information between words and visual information, affects word-by-word reading times.

事实上，预测 1 已经由 Pimentel 等人（2023）进行了研究，他们的结果证实了我们的预测。Pimentel 等人将给定一组先前词语的情况下，下一个词语的熵称为一个词的上下文熵（contextual entropy）。他们发现，随着词级别的上下文熵增加，阅读时间也随之增加。因此，在本文的其余部分，我们感兴趣的是测试预测 2，即词语与视觉信息之间的互信息作为自下而上证据质量的模型，是否会影响逐词阅读时间。

3 Methods

3 方法

3.1 Materials

3.1 材料

We use a portion of the advanced OneStopQA dataset (Berzak et al., 2020). This dataset contains Guardian news articles, along with carefully constructed reading comprehension questions, which are linked to individual spans in the text. For our study, we selected three articles: “101-Year-Old Bottle Message”, “Inky the Octopus Escapes from Aquarium”, and “Japan Calls Time on Long Hours Work Culture”. A team member with experience in English-Chinese translation hand-translated these texts and their questions into Mandarin. This small

我们使用了先进的一站式问答数据集（Berzak 等，2020）的一部分。该数据集包含《卫报》新闻文章，以及精心构建的阅读理解问题，这些问题与文本中的特定片段相关联。在我们的研究中，我们选择了三篇文章：“101 岁瓶中信”、“章鱼 ‘inky’ 从水族馆逃出”以及“日本叫停超时工作文化”。一名具有英汉翻译经验的团队成员手译了这些文本及其问题为中文。这个小型



Figure 2: images/90e145bceea742d5a4c0da135f9ce5fff31ef9daa233b1af27bc8226e34b1b18.jpg

Figure 1: Example showing a screen from a MoTR trial with our three different reading conditions.

图 1：示例展示了一个 MoTR 试验中的屏幕，包含我们三种不同的阅读条件。

translated corpus, which we term the Chinese OneStopQA, is released along with the publication of this article (see code repository).

翻译后的语料库，我们称之为中文 OneStopQA，将与本文的发布一同公布（参见代码仓库）。

The English subset contains 1,793 words (mean word length = 4.6, SD = 2.53), while the Chinese subset contains 3,182 characters. In terms of experimental presentation, one Chinese character occupies roughly 1.46 times the pixel space of an English letter, making an average English word about 3.2 times longer than a Chinese character. The average Zipf frequency is slightly higher in English ($M = 5.77$, $SD = 1.45$) than in Chinese ($M = 4.84$, $SD = 1.90$), largely due to the low frequency of transliterated Western named entities in the Chinese translations.

英文子集包含 1,793 个单词（平均单词长度 = 4.6，标准差 = 2.53），而中文子集包含 3,182 个字符。从实验呈现的角度来看，一个中文字符占据的像素空间大约是英文字符的 1.46 倍，因此一个平均英文单词的长度大约是中文字符的 3.2 倍。在 Zipf 频率方面，英文的平均频率略高于中文（英文 $M=5.77$ ，标准差 = 1.45；中文 $M=4.84$ ，标准差 = 1.90），这主要是由于中文翻译中西方专有名词的音译词频率较低所致。

Creating Noisy Words To create noised reading conditions, we occluded (i.e., masked with white) either the upper or lower half of every word in the dataset. There are potentially many ways to add noise to the texts. Other options would be to occlude the first half or the second half of words,

as well as Gaussian noise. Previous work has shown that the beginnings of words tend to carry more disambiguatory information than their endings. For example, Pimentel et al. (2021) demonstrates this cross-linguistically using information-theoretic measures, while Alhama et al. (2019) presents a perceptually constrained connectionist model explaining fixation biases toward word onsets. These findings are consistent with psycholinguistic evidence of the optimal viewing position effect in visual word recognition (Brysbaert and Nazir, 2005).

创建噪声文字为了创建有噪声的阅读条件，我们将数据集中每个单词的上半部分或下半部分遮挡（即用白色遮盖）。向文本添加噪声的方式可能有很多种。其他选项包括遮挡单词的第一部分或第二部分，以及添加高斯噪声。先前的研究表明，单词的开头往往比结尾携带更多的消歧信息。例如，Pimentel 等人（2021）使用信息论度量跨语言地展示了这一点，而 Alhama 等人（2019）则提出了一种感知约束的连接主义模型，解释了人们倾向于注视单词开头的偏倚。这些发现与视觉单词识别中最佳视点效应的心理语言学证据一致（Brysbaert 和 Nazir, 2005）。

However, these studies focus on the linear distribution of information across letter positions, which applies naturally to alphabetic scripts such as English but not to logographic systems like Chinese, where characters are two-dimensional and not arranged linearly. We were also concerned that completely removing some letters or characters would make reading too difficult or frustrating for participants, and that the removal of letters or characters demands very careful handling to avoid confounds (Rayner, 1998; Rayner et al., 2006). Masking the upper or lower half retains some information about each character, which presents a paradigm that is relatively readable for participants, especially in the degraded conditions. In addition, unlike simply adding Gaussian noise, upper and lower half occlusion allows us to investigate where, in vertical space, information is localized in English and Chinese orthographic systems. Our strategy leads to two additional research questions:

然而，这些研究关注的是信息在字母位置上的线性分布，这在如英语等字母文字中是自然适用的，但在如中文等表意文字中则不适用，因为汉字是二维的，并非线性排列的。我们还担心，完全移除某些字母或汉字会使参与者的阅读变得过于困难或令人沮丧，而且移除字母或汉字需要非常谨慎地处理，以避免混淆因素（Rayner, 1998; Rayner et al., 2006）。遮蔽上半部分或下半部分保留了每个字符的一些信息，这为参与者提供了一个相对易于阅读的范式，尤其是在退化条件下。此外，与简单地添加高斯噪声不同，上半部分和下半部分遮蔽使我们能够研究在英语和中文文字系统中，信息在垂直空间中的定位位置。我们的策略导致了两个额外的研究问题：

Sub Research Question 1 Is information split up differentially between the upper and lower halves of orthographic words?

子研究问题 1 拼写词的上半部分和下半部分之间信息是否被差异性地分割？

Sub Research Question 2 Does the relative informativeness of upper vs. lower halves differ across languages?

子研究问题 2 上下半部分的相对信息量在不同语言中是否存在差异？

3.2 Data Collection

3.2 数据收集

Mouse Tracking for Reading (MoTR) To test our main predictions, we need a way of measuring (average) human reading times in our different conditions. To do so, we use Mouse Tracking for Reading (MoTR; Wilcox et al., 2024). In a MoTR trial, a blurred text is presented on a screen. A small region around the tip of a user's mouse brings the text into focus. Participants move the mouse to incrementally reveal and read the text, while their mouse location is recorded and used as a proxy for gaze location. The time-stamped x/y coordinates are then turned into incremental word-by-word reading times, similar to word-level reading times in an eye-tracking-while-reading experiment. As in eye-tracking, there are several ways to compute reading times. For our main analysis, we use the first-pass reveal time (FPRT), defined as the total amount of time

a participant spends revealing a word during their first pass reading. Conveniently, the same acronym (FPRT) is used in eye-tracking for first-pass reading time, but we use “reveal” to emphasize that it is computed from mouse movements rather than eye fixations.

鼠标追踪阅读 (MoTR) 为了检验我们的主要预测, 我们需要一种方法来测量 (平均) 人类在不同条件下的阅读时间。为此, 我们使用鼠标追踪阅读 (MoTR; Wilcox 等, 2024)。在一个 MoTR 实验中, 屏幕上会显示一段模糊的文本。用户鼠标尖端附近的小区域会使文本变得清晰。参与者移动鼠标以逐步揭示并阅读文本, 同时记录鼠标的移动位置, 并将其作为注视位置的代理。随后, 带有时间戳的 x/y 坐标会被转换为逐字的阅读时间, 类似于在眼动追踪阅读实验中的逐字阅读时间。与眼动追踪类似, 计算阅读时间有多种方法。在我们主要的分析中, 我们使用首次通过揭示时间 (FPRT), 定义为参与者在首次通读时揭示每个单词所花费的总时间。方便的是, 眼动追踪中也使用相同的缩写 (FPRT) 表示首次通过阅读时间, 但我们使用 “揭示” 一词来强调它是基于鼠标的移动而非眼动固定来计算的。

Wilcox et al. (2024) show that MoTR reading measures are strongly correlated with eye-tracking and self-paced-reading measures. MoTR has been used to collect data in English and Russian (Oğuz et al., 2025), but not in Chinese.

Wilcox 等人 (2024) 表明, MoTR 阅读测量与眼动追踪和自主阅读测量有很强的相关性。MoTR 已被用于在英语和俄语中收集数据 (Oğuz 等人, 2025), 但尚未在中国使用。

Participants We recruited 54 English and 57 Chinese speakers on Prolific, requiring a minimum approval rate of 98% and the corresponding language to be their first and native language. Participants were compensated 3.75 GBP for a median reading time of 25 minutes.

参与者我们在 Prolific 上招募了 54 名英语母语者和 57 名中文母语者, 要求其最低批准率为 98%, 且相应语言为其第一语言和母语。参与者获得 3.75 英镑的报酬, 平均阅读时间为 25 分钟。

Procedure Each participant read the article paragraphs presented screen by screen, with each screen randomly assigned to one of three conditions: upper-half occluded (i.e., lower-half visible), lower-half occluded (i.e., upper-half visible), or unoccluded (see Figure 1). In addition to reading texts and answering comprehension questions, we ask participants to rate the ease of reading after finishing all the trials using a multiple-choice question: “Which do you find easier to read: text showing only the top half or only the bottom half?” The options are “Top half only,” “Bottom half only,” and “About the same.” The actual experiments are available online for Chinese⁴ and English⁵.

程序每位参与者逐屏阅读文章段落, 每屏随机分配至三种条件之一: 上半部遮挡 (即下半部可见)、下半部遮挡 (即上半部可见) 或无遮挡 (见图 1)。除了阅读文本和回答理解问题外, 我们还要求参与者在完成所有试验后, 使用多项选择题评估阅读的难易程度: “您认为哪种文本更容易阅读: 仅显示上半部还是仅显示下半部?” 选项包括 “仅上半部”、“仅下半部” 和 “大致相同”。实际实验可在线获取, 中文⁴ 和英文⁵。

3.3 Mutual Information Estimation

3.3 相互信息估计

In Section 2, our model concerns the mutual information between words, W , and (visual) evidence sampled by the reader, E . However, we do not have direct access to this evidence. Instead, as a proxy for our visual evidence, we estimate the mutual information between words W and their orthographic representations $o \in R^d$, where o is a realization of a random variable O that ranges over representations of different words. Following Pimentel et al. (2020), we decompose the mutual information as

在第 2 节中, 我们的模型关注的是词语 W 与读者采样的 (视觉) 证据 E 之间的互信息。然而, 我们无法直接获取这种证据。相反, 作为我们视觉证据的代理, 我们估计词语 W 与其拼写表示 $PH0000$ 之间的互信息, 其中 o 是一个随机变量 O 的实现, O 的取值范围是不同词语的表示形式。根据 Pimentel 等人 (2020) 的研究, 我们将互信息分解为

$$I(W; \mathbf{o}) = H(W) - H(W | \mathbf{o}) \quad (8a)$$

$$\approx H_\theta(W) - H_\theta(W | \mathbf{o}) \quad (8b)$$

where θ denotes the parameters of the models employed for entropy estimation. We estimate each of these two terms separately.

其中 θ 表示用于熵估计的模型的参数。我们分别估计这两个项。

We estimate unconditional entropy $H_\theta(W)$ with a maximum likelihood estimation of the uni-gram distribution of Chinese characters and English words. We take the 9,933 unique Chinese characters included in the modern Chinese character database⁶, and the 60,384 English words in the SUBTLEXus database (Brysbaert and New, 2024), and look up their frequencies using the Python library wordfreq (Speer, 2022) that supports both languages and aggregates data from multiple domains, including subtitles, Wikipedia, news, fiction, and web content. Normalizing the frequencies, we obtain the empirical distribution $p_\theta(w)$ and from it we can directly compute the entropy $H_\theta(W)$. The empirical entropies are 5.59 and 7.12 nats for Chinese characters and English words.

我们使用中文字符和英文单词的单字分布的最大似然估计来估计无条件熵 $H_\theta(W)$ 。我们采用现代汉字数据库⁶中包含的 9,933 个独特汉字，以及 SUBTLEXus 数据库 (Brysbaert 和 New, 2024) 中的 60,384 个英文单词，并使用支持中英文且从多个领域（包括字幕、维基百科、新闻、小说和网络内容）聚合数据的 Python 库 wordfreq (Speer, 2022) 查找它们的频率。对频率进行归一化处理后，我们得到经验分布 $p_\theta(w)$ ，从而可以直接计算熵 $H_\theta(W)$ 。经验熵值分别为汉字 5.59 nats 和英文单词 7.12 nats。

We estimate the conditional entropy $H_\theta(W | \mathbf{o})$ in two stages. First, we compute the word entropy conditioned on a specific orthographic representation, $H_\theta(W | \mathbf{o} = \mathbf{o})$. We refer to this as pointwise conditional entropy. We compute this value by taking the expectation of the information content, or surprisal of all words given the orthographic representation $\iota_\theta(w | \mathbf{o})$, where $\iota_\theta(\cdot) = -\log p_\theta(\cdot)$. Given a model with parameters θ that can produce our probability distribution of interest, that is, $p_\theta(w | \mathbf{o})$, the pointwise conditional entropy is calculated as:

我们分两个阶段估计条件熵 $H_\theta(W | \mathbf{o})$ 。首先，我们计算在特定拼写表示 $H_\theta(W | \mathbf{o} = \mathbf{o})$ 下的词熵。我们将这一过程称为点条件熵。我们通过计算所有词语在给定拼写表示 $\iota_\theta(w | \mathbf{o})$ 时的信息内容，或意外性 (surprisal) 的期望值来计算该值，其中 $\iota_\theta(\cdot) = -\log p_\theta(\cdot)$ 。给定一个具有参数 θ 的模型，该模型可以生成我们感兴趣的概率分布，即 $p_\theta(w | \mathbf{o})$ ，点条件熵的计算公式为：

$$H_\theta(W | \mathbf{o}) \approx \sum_{w \in \mathcal{W}} p_\theta(w | \mathbf{o}) \iota_\theta(w | \mathbf{o}) \quad (9)$$

We then estimate conditional entropy as the expectation of the pointwise conditional entropy with respect to $\mathbf{o}$, following the identity $H(W | \mathbf{o}) = \mathbb{E}_{\mathbf{o}}[H(W | \mathbf{o} = \mathbf{o})]$. We take the expectation over a set of held-out test samples:

我们随后根据恒等式 $H(W | \mathbf{o}) = \mathbb{E}_{\mathbf{o}}[H(W | \mathbf{o} = \mathbf{o})]$ ，将条件熵估计为关于 $\mathbf{o}$ 的点条件熵的期望值。我们对一组保留的测试样本进行期望计算：

$$H_\theta(W | \mathbf{o}) \approx \frac{1}{N} \sum_{n=1}^N H_\theta(W | \mathbf{o}^n) \quad (10)$$

where $\mathbf{o}^n$ is the n^{th} orthographic representation in the test set.

其中 o^n 是测试集中的 n^{th} 的正写法表示。

We note that using these methods, we can estimate not only the mutual information $I(W; \mathbf{o})$, but also its pointwise variant, also called the information gain (IG), for a particular orthographic representation, where $IG(W; \mathbf{o}) = H(W) - H(W | \mathbf{o})$. While our formal prediction is made in terms of mutual information, in Section 4.3, we use IG to investigate the relationship between information contained in individual visual inputs and their respective reading times.

我们注意到，使用这些方法，我们不仅可以估计互信息 $I(W; \mathbf{o})$ ，还可以估计其点态变体，也称为信息增益 (IG)，用于特定的拼写表示，其中 $IG(W; \mathbf{o}) = H(W) - H(W | \mathbf{o})$ 。虽然我们的正式预测是以互信息为基础的，但在第 4.3 节中，我们使用 IG 来研究单个视觉输入中所包含的信息与其相应的阅读时间之间的关系。

In recent work, similar methods have been used to study the relationship between words (as represented by text) and prosody, or the melody of speech (Wolf et al., 2023; Regev et al., 2025; Wilcox et al., 2025). However, these previous works learn distributions over real-valued variables that represent pitch. We wish to learn distributions over discrete w-valued variables $p_\theta(w | \mathbf{o})$. To obtain this distribution, we use multimodal language

在最近的工作中，类似的方法已被用于研究词语（以文本形式表示）与语调（即言语的旋律）之间的关系 (Wolf 等, 2023; Regev 等, 2025; Wilcox 等, 2025)。然而，这些先前的研究学习的是表示音高的实值变量上的分布。我们希望学习的是在离散的 w 值变量上的分布 $p_\theta(w | \mathbf{o})$ 。为了获得这种分布，我们使用多模态语言

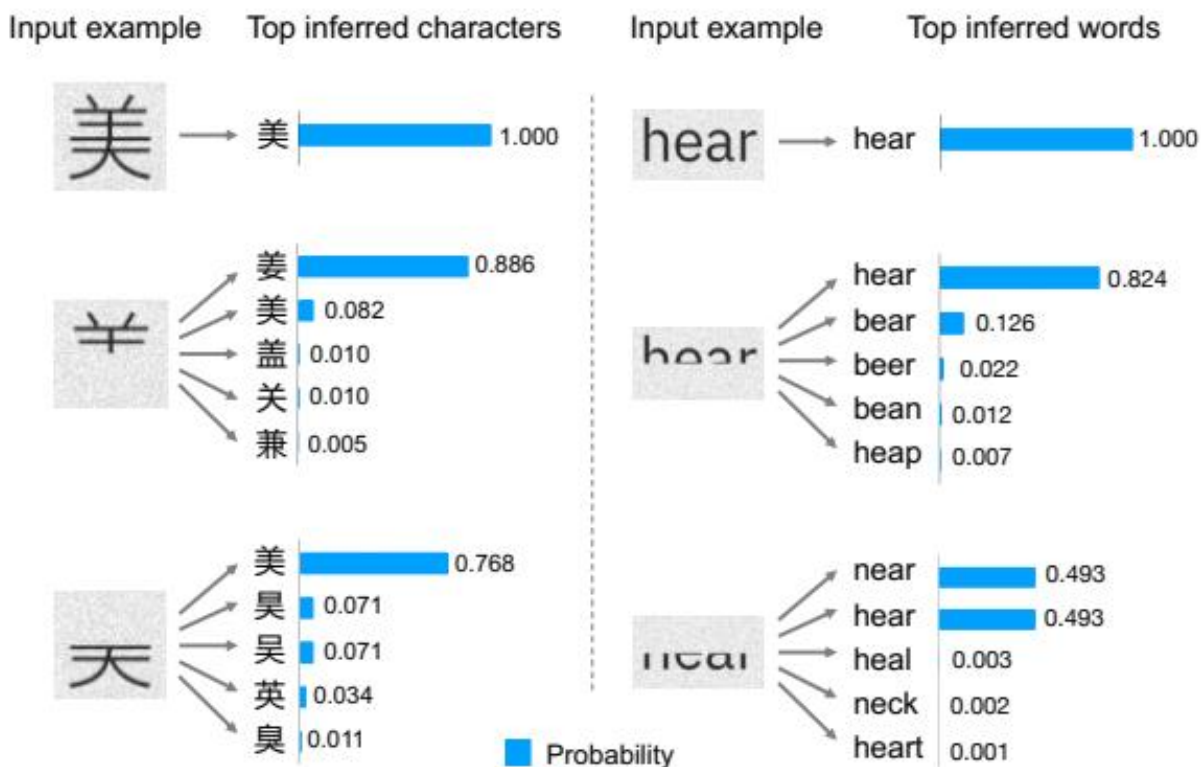


Figure 3: images/21056261e599757a53bbb37d704265e891642615f5ff1e1fd65390ab26308f71.jpg

Figure 2: Results of fine-tuned Qwen2.5 model for the Chinese character 美 (“beautiful”) and the English word hear. The preference for hear over heal in upper half occlusion likely reflects pre-training frequency bias, which we control for by training TransOCR from scratch.

图 2：微调后的 Qwen2.5 模型对汉字“美” (“beautiful”) 和英文单词 hear 的结果。在上半部分遮挡情

况下，对 hear 的偏好可能反映了预训练频率偏差，我们通过从头开始训练 TransOCR 来控制这一偏差。

models, which we fine-tune to produce conditionalized distributions over words, given visual inputs. We do so with the following methods:

模型，我们对其进行微调，以根据视觉输入生成条件化的词分布。我们采用以下方法：

Data Generation We adapt the Python library *TRDG*⁷ to generate images of Chinese characters and English words from text as their orthographic representations, applying upper-, lower-half occlusion to create our different experimental conditions. For each character or word under each condition, we generated six images with different fonts and font weights⁸ to enhance visual variability, and added a small amount of Gaussian noise to the image backgrounds (Li et al., 2025). We generate 16,800 Chinese character images and 44,800 English word images under each of the three occlusion conditions as training data. For test data, we generate images of all Chinese characters in the Chinese OneStopQA dataset and all English words in the selected OneStopQA subset, again under each occlusion condition.

数据生成我们采用 Python 库 *TRDG*⁷，将文本转换为汉字和英文单词的字形表示，通过上半部和下半部遮挡来创建不同的实验条件。对于每个条件下的每个字符或单词，我们生成六张使用不同字体和字重的图像⁸以增强视觉多样性，并在图像背景中添加少量高斯噪声 (Li 等, 2025)。在每种遮挡条件下，我们生成 16,800 张汉字图像和 44,800 张英文单词图像作为训练数据。对于测试数据，我们生成中国 OneStopQA 数据集集中的所有汉字和所选 OneStopQA 子集中的所有英文单词的图像，同样在每种遮挡条件下进行生成。

A Bayesian Baseline Model As a simple reference, we implement a Bayesian baseline to estimate the pointwise conditional entropy $H(W \mid \mathbf{O} = \mathbf{o})$. In this model, lexical frequencies provide priors $p(w)$, and structural similarity (SSIM; Wang et al., 2004), computed with scikit-image (Van der Walt et al., 2014), serves as a likelihood $p(\mathbf{o} \mid w)$. Posterior probabilities $p(w \mid \mathbf{o})$ are obtained by normalizing the product of priors and likelihoods across

一个贝叶斯基线模型作为简单的参考，我们实现了一个贝叶斯基线模型来估计逐点条件熵 $H(W \mid \mathbf{O} = \mathbf{o})$ 。在该模型中，词汇频率提供先验 $p(w)$ ，而结构相似性 (SSIM; Wang 等人, 2004)，使用 scikit-image (Van der Walt 等人, 2014) 计算，作为似然 $p(\mathbf{o} \mid w)$ 。通过将先验和似然的乘积在跨... 上归一化，得到后验概率 $p(w \mid \mathbf{o})$ 。

all candidate images, with the denominator corresponding to the marginal likelihood $p(\mathbf{o})$. We then compute $H(W \mid \mathbf{O} = \mathbf{o})$ from these posteriors for each input image and average across the dataset to obtain $H(W \mid \mathbf{O})$. While straightforward, this baseline has some limitations. First, its computational cost scales quadratically with dataset size ($O(N^2)$). Second, SSIM captures only low-level pixel similarity, often producing clustered scores for orthographically distinct characters or words. Third, its estimates are largely driven by lexical frequency. For these reasons, we turn to multimodal models based on artificial neural networks (ANNs) for more reliable and scalable estimation.

所有候选图像，其中分母对应于边缘似然 $p(\mathbf{o})$ 。然后，我们从这些后验分布中计算每个输入图像的 $H(W \mid \mathbf{O} = \mathbf{o})$ ，并在整个数据集上进行平均以获得 $H(W \mid \mathbf{O})$ 。虽然这种方法简单，但该基线方法有一些局限性。首先，其计算成本随着数据集大小呈二次增长 ($O(N^2)$)。其次，SSIM 仅捕捉低层次的像素相似性，常常会对正交不同的字符或单词产生聚集的分数。第三，其估计结果在很大程度上受到词频的影响。因此，出于这些原因，我们转向基于人工神经网络 (ANNs) 的多模态模型，以实现更可靠和可扩展的估计。

ANN-based Predictive Multimodal Models We use three different multimodal model settings: First, we evaluate the pre-trained Qwen2.5-VL-7B-Instruct⁹ in a zero-shot setting. Qwen2.5-VL-7B is an open-source vision-language model developed by Alibaba, designed for high-accuracy multimodal analysis with enhanced visual understanding and text-image alignment (Wang et al., 2024; Bai et al., 2025). As upper- and lower-half occluded words are likely out-of-distribution with respect to the model’s training data, we do not expect the mutual information estimate to be tight in this setting. For a better estimate, we then fine-tune Qwen2.5-VL-7B on our task-specific

data to improve its performance. To complement the estimate from the pre-trained model, we also train a separate transformer-based OCR model (TransOCR; Yu et al., 2023), from scratch, to perform the same prediction task. The model combines a ResNet encoder with a Transformer decoder for character recognition. Full training configurations and prompt designs for the Qwen and TransOCR models are provided in Section B and Section C, respectively.

基于 ANN 的预测多模态模型我们使用了三种不同的多模态模型设置：首先，我们在零样本设置下评估预训练的 Qwen2.5-VL-7B-Instruct⁹。Qwen2.5-VL-7B 是由阿里巴巴开发的开源视觉-语言模型，旨在实现高精度的多模态分析，具有增强的视觉理解和文本-图像对齐功能（Wang 等，2024；Bai 等，2025）。由于上下半部遮挡的词语可能与模型的训练数据不一致，因此我们不期望在此设置下互信息估计能够紧密。为了获得更好的估计，我们随后在我们的任务特定数据上对 Qwen2.5-VL-7B 进行微调，以提高其性能。为了补充预训练模型的估计，我们还从头开始训练了一个基于 Transformer 的 OCR 模型（TransOCR; Yu 等，2023），以执行相同的预测任务。该模型结合了 ResNet 编码器和 Transformer 解码器进行字符识别。Qwen 和 TransOCR 模型的完整训练配置和提示设计分别在第 B 节和第 C 节中提供。

4 Results

4 结果

4.1 Human Reading Results

4.1 人工阅读结果

We show reading times in Figure 3(a). In both languages, reading full words resulted in the shortest average FPRT, as predicted. Interestingly, both languages follow a Full < Upper < Lower pattern, with lower-half visibility leading to the longest FPRT. To quantify these effects, we fit linear mixed-effects models with visibility condition as a predictor, using sliding contrasts to compare Upper vs. Full and

我们在图 3(a) 中展示了阅读时间。在两种语言中，阅读完整单词导致的平均 FPRT 最短，正如预测的那样。有趣的是，两种语言都遵循“完整 < 上半部分 < 下半部分”的模式，下半部分的可见性导致 FPRT 最长。为了量化这些效应，我们使用滑动对比方法，将上半部分与完整进行比较，以可见性条件作为预测变量来拟合线性混合效应模型，

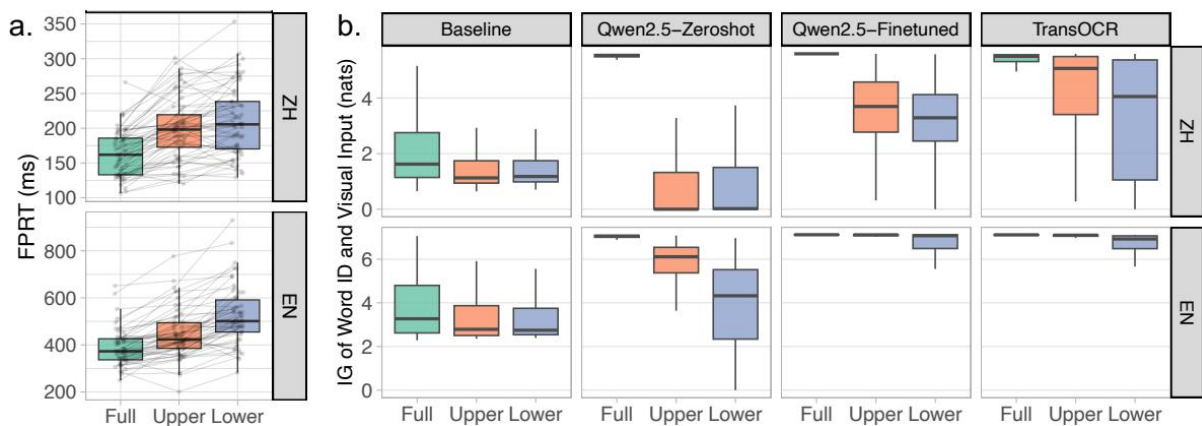


Figure 3: (a) Reading times (FPRT) measured under three visibility conditions. Boxes represent the interquartile range (middle 50%), center lines indicate the median, and whiskers show the overall data spread. Grey lines trace each participant’s mean across conditions. EN: English; ZH: Simplified Chinese (b) Information gain (IG) between word identity and visual form under the three conditions, obtained with each of our estimation techniques.

图 3: (a) 在三种可见度条件下测量的阅读时间 (FPRT)。方框表示四分位数范围 (中间 50%)，中心线表示中位数，须须线显示整体数据分布。灰色线条追踪每位参与者在不同条件下的平均值。EN：英语；ZH：中文。

简体中文 (b) 在三种条件下，单词身份与视觉形式之间的信息增益 (IG)，采用我们每种估计技术获得。

Lower vs. Upper. Word length (EN only), lexical frequency, surprisal, and contextual entropy are included as control variables, with random intercepts for subjects and items. In Chinese, both contrasts are significant: $\beta = 37.79$ ms ($p < 2e-16$) and $\beta = 12.64$ ms ($p < 2e-16$). In English, the effects are larger: $\beta = 69.93$ ms ($p < 2e-16$) and $\beta = 90.09$ ms ($p < 2e-16$)¹⁰.

下 vs. 上。词长 (仅限英文)、词汇频率、意外性以及上下文熵被作为控制变量，且对被试和项目设置了随机截距。在中文中，两个对比均显著： $\beta = 37.79$ 毫秒 ($p < 2e-16$) 和 $\beta = 12.64$ 毫秒 ($p < 2e-16$)。在英语中，效应更大： $\beta = 69.93$ 毫秒 ($p < 2e-16$) 和 $\beta = 90.09$ 毫秒 ($p < 2e-16$)¹⁰。

These results can be interpreted as implying a visual asymmetry in both languages between ease of processing with respect to just upper and lower halves of words. The asymmetry is stronger in English, where the lower half leads to greater slowdowns. Participants' subjective ratings confirm this asymmetric pattern and further show that English lower halves are perceived as harder to read than Chinese ones (Appendix D). In addition, we summarize comprehension question performance in Appendix E. Accuracy declines with occlusion, but remains well above chance (25%), indicating that reading is effective.

这些结果可以被解释为表明，在两种语言中，词语上半部分和下半部分的处理难度存在视觉不对称性。这种不对称性在英语中更为明显，其中下半部分会导致更大的阅读速度减慢。参与者主观的评分确认了这种不对称模式，并进一步表明，英语词语的下半部分被认为比中文词语的下半部分更难阅读 (附录 D)。此外，我们还在附录 E 中总结了理解问题的答题表现。准确性随着遮挡程度的增加而下降，但仍然远高于随机水平 (25%)，这表明阅读是有效的。

4.2 Mutual Information Results

4.2 互信息结果

To give a visual sense of how our models perform, in Figure 2, we show sample images in the three experimental conditions, along with the predictions from the fine-tuned Qwen2.5 model. As a performance check, we also report the model accuracies (Acc) in Table 1. Baseline accuracies are not reported because they only compare each test image against other images in the test set, rather than predicting over the full vocabulary. This makes its ac-

为了直观地展示我们的模型表现，我们在图 2 中展示了三种实验条件下的示例图像，以及经过微调的 Qwen2.5 模型的预测结果。作为性能检查，我们在表 1 中报告了模型的准确率 (Acc)。未报告基线准确率，因为它们仅将每个测试图像与其他测试集中的图像进行比较，而不是对整个词汇表进行预测。这使得其准确率计算方式有所不同。

Language	Model	Acc (%)			MI (nat)		
Full	Upper	Lower	Full	Upper	Lower		
ZH	Baseline	-	-	-	4.85	4.42	4.41
Qwen2.5-zs	97.9	5.2	3.2	5.42	0.27	0.32	
Qwen2.5-ft	99.0	48.6	44.3	5.57	3.62	3.27	
TransOCR	97.9	78.8	65.7	5.26	4.09	3.17	
EN	Baseline	-	-	-	6.43	6.12	5.99
Qwen2.5-zs	98.8	84.1	51.5	6.99	5.74	3.86	
Qwen2.5-ft	99.7	93.1	68.1	7.11	7.01	6.66	
TransOCR	98.8	95.7	65.8	7.07	7.00	6.68	

语言	模型	准确率 (%)		MI (自然)				
		上部	下层	完整	上部	下部		
完整								
请提供需要翻译的文本内容。	基线	-	-	-	4.85	4.42	4.41	
Qwen2.5-zs	97.9	5.2	3.2	5.42	0.27	0.32		
Qwen2.5-ft	99.0	48.6	44.3	5.57	3.62	3.27		
TransOCR	97.9	78.8	65.7	5.26	4.09	3.17		
EN	基线	-	-	-	6.43	6.12	5.99	
Qwen2.5-zs	98.8	84.1	51.5	6.99	5.74	3.86		
Qwen2.5-ft	99.7	93.1	68.1	7.11	7.01	6.66		
TransOCR	98.8	95.7	65.8	7.07	7.00	6.68		

Table 1: Model accuracy (%) and mutual information estimates, $I(W; \mathbf{O})$, for Chinese (ZH) and English (EN) with different models.

表 1: 模型准确率 (%) 和互信息估计值, $I(W; \mathbf{O})$, 对于中文 (ZH) 和英文 (EN) 使用不同模型的结果。

accuracy values not directly comparable to the other models. Accuracies in the unoccluded Full condition are uniformly high ($> 97\%$). Under occlusion, zero-shot Qwen2.5 drops dramatically in Chinese ($< 6\%$) but is still strong in English (52–85%). Fine-tuning improves performance in both languages, while TransOCR achieves the most robust accuracy overall.

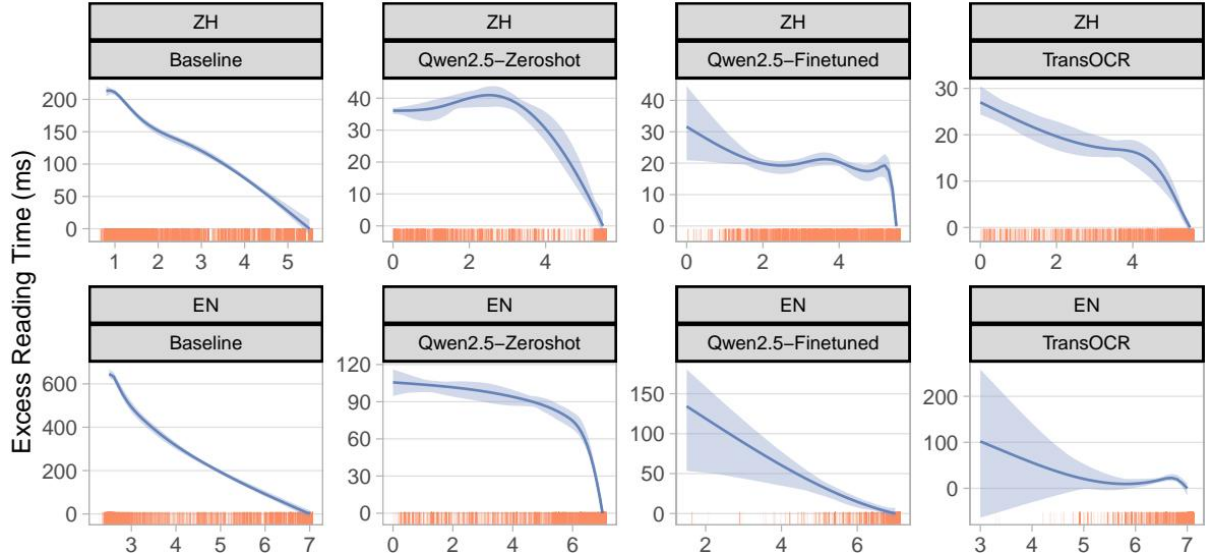
准确性值不能直接与其他模型进行比较。在未遮挡的完整条件下, 准确率普遍较高 ($> 97\%$)。在遮挡情况下, 零样本 Qwen2.5 在中文中大幅下降 ($< 6\%$), 但在英文中仍保持强劲 (52–85%)。微调在两种语言中均提升了性能, 而 TransOCR 在整体上实现了最稳健的准确率。

Our main analyses focus on mutual information, $I(W; O)$ in Table 1 and information gain, $IG(W; o)$ in Figure 3(b), across conditions and models. Figure 3(b) shows IG between word identity and visual input under three visibility conditions, estimated with the Bayesian baseline, Qwen2.5-VL-7B-Instruct (zero-shot and fine-tuned), and TransOCR. Table 1 reports MI as the IG averaged across all inputs. MI decreases systematically from Full to Upper to Lower in both languages.

我们的主要分析集中在互信息 (Mutual Information) 上, 如表 1 所示, 以及在不同条件和模型下的信息增益 (Information Gain), 如图 3(b) 所示。图 3(b) 展示了在三种可见性条件下, 单词身份与视觉输入之间的信息增益 (IG), 使用贝叶斯基线模型 Qwen2.5-VL-7B-Instruct (零样本和微调) 和 TransOCR 进行估计。表 1 报告了 MI 作为所有输入的平均信息增益。在两种语言中, MI 从 Full 逐渐降低到 Upper 再到 Lower。

For a statistical test of our observed trends, we fit linear mixed-effects models for each language-model pair, with visibility as the main predic-

为了对我们的观察到的趋势进行统计检验, 我们为每对语言模型拟合线性混合效应模型, 以可见性作为主要预测变量。



IG of Word Identity and Visual Input (nats)

Figure 4: Relationship between informational quality of individual words (information gain; IG) and excess reading time. Solid blue lines are smoothed GAM fits; shaded regions show bootstrapped 95% confidence intervals. Red tick marks along the bottom (rug plots) indicate the distribution of IG data points. Reading times are aligned to end at zero at the highest MI end to emphasize the relative excess reading time when information quality decreases.

词的识别身份与视觉输入 (nats)

图 4：单个词的信息质量（信息增益；IG）与额外阅读时间之间的关系。实线蓝色线条为平滑的 GAM 拟合曲线；阴影区域显示的是通过自助法得到的 95% 置信区间。底部的红色刻度标记（rug plots）表示 IG 数据点的分布。在最高 MI 端，阅读时间被对齐至零，以强调当信息质量下降时的相对额外阅读时间。

tor. As in the reading time analysis, we use the same sliding contrasts. Word length (EN only), lexical frequency, surprisal, and contextual entropy are included as controls, with random intercepts for items.¹¹ In Chinese, all models show significant IG reductions when only the upper half is visible (Baseline: $\beta = -.58$; Qwen2.5-Zeroshot: $\beta = -4.55$; Qwen2.5-Finetuned: $\beta = -1.85$; TransOCR: $\beta = -.99$ nats), and IG from fine-tuned models dropped further when visibility changed from Upper to Lower (Qwen2.5-Finetuned: $\beta = -.37$; TransOCR: $\beta = -1.01$ nats). In English, the zero-shot model showed the largest overall drop (Upper vs. Full: $\beta = -1.43$; Lower vs. Upper: $\beta = -2.09$ nats), while the baseline and fine-tuned models show smaller but consistent reductions (Baseline: $\beta = -.36, -.07$; Qwen2.5-Finetuned: $\beta = -.11, -.47$; TransOCR: $\beta = -.08, -.35$ nats). All effects were statistically significant at $p < .0001$. Panels (a) and (b) of Figure 3, taken together, reveal a clear pattern: as visual input degrades from Full to Upper to Lower, as measured by IG, reading times increase.

正如在阅读时间分析中所使用的，我们采用了相同的滑动对比。控制变量包括单词长度（仅限英文）、词频、惊讶度和上下文熵，且对项目使用随机截距。¹¹ 在中文中，所有模型在仅上半部分可见时都显示出显著的 IG 下降（基线： $\beta = -.58$ ；Qwen2.5-Zeroshot： $\beta = -4.55$ ；Qwen2.5-Finetuned： $\beta = -1.85$ ；TransOCR： $\beta = -.99$ nats），当可见性从上半部分变为下半部分时，微调模型的 IG 进一步下降（Qwen2.5-Finetuned： $\beta = -.37$ ；TransOCR： $\beta = -1.01$ nats）。在英文中，零样本模型显示出最大的总体下降（上半部分 vs. 完整： $\beta = -1.43$ ；下半部分 vs. 上半部分： $\beta = -2.09$ nats），而基线模型和微调模型显示出较小但一致的下降（基线： $\beta = -.36, -.07$ ；Qwen2.5-Finetuned： $\beta = -.11, -.47$ ；TransOCR： $\beta = -.08, -.35$ nats）。所有效应在 $p < .0001$ 处均具有统计学显著性。图 3 的 (a) 和 (b) 部分共同揭示了一个清晰的模式：随着视觉输入从完整变为上半部分再变为下半部分，如通过 IG 衡量的，阅读时间随之增加。

4.3 Word-Level Relationship

4.3 单词级关系

In this section, we test the relationship between reading time and informational quality at the word level. To do so, we fit linear mixed-effects models with reading time of an orthographic representation as the dependent variable and its IG as a fixed effect. We also included frequency, surprisal, contextual entropy, and (in EN) word length as additional fixed effects, as well as by-subject and by-item random intercepts.

在本节中，我们测试了在单词层面阅读时间与信息质量之间的关系。为此，我们将阅读时间作为因变量，以字形表示的 IG 作为固定效应，拟合线性混合效应模型。我们还加入了词频、意外性、上下文熵以及（在英语中）词长作为额外的固定效应，同时还包含了被试间和项目间的随机截距。

We find a significant effect of IG on reading time across all models and measures, with a consistent negative effect: the higher the informational quality of the input, the faster it is read. In Chinese, all four IG estimates were significant predictors of FPRT: $\beta = -14.94$ ms (Baseline), $\beta = -7.53$ ms (Qwen2.5-Zeroshot), $\beta = -10.19$ ms (Qwen2.5-Finetuned), and $\beta = -4.97$ ms (TransOCR). In English, the effects were even larger: $\beta = -15.36$ ms (Baseline), $\beta = -23.67$ ms (Qwen2.5-Zeroshot), $\beta = -51.48$ ms (Qwen2.5-Finetuned), and $\beta = -66.42$ ms (TransOCR). All effects were statistically significant at $p < .0001$.

我们在所有模型和测量方法中都发现了信息质量（IG）对阅读时间的显著影响，且影响具有一致的负向性：输入的信息质量越高，阅读速度越快。在中文中，所有四个 IG 估计值都是 FPRT 的显著预测因子： $\beta = -14.94$ 毫秒（基线）， $\beta = -7.53$ 毫秒（Qwen2.5-Zeroshot）， $\beta = -10.19$ 毫秒（Qwen2.5-Finetuned），以及 $\beta = -4.97$ 毫秒（TransOCR）。在英文中，效果甚至更大： $\beta = -15.36$ 毫秒（基线）， $\beta = -23.67$ 毫秒（Qwen2.5-Zeroshot）， $\beta = -51.48$ 毫秒（Qwen2.5-Finetuned），以及 $\beta = -66.42$ 毫秒（TransOCR）。所有效果在统计学上均显著（ $p < .0001$ ）。

4.4 Nonlinear Relationship Between Information Quality and Reading Time

4.4 信息质量与阅读时间之间的非线性关系

While our linear regression models show that informational quality affects reading time, it makes strong assumptions about the functional form of this relationship. In order to get a better sense of

虽然我们的线性回归模型显示信息质量影响阅读时间，但它对这种关系的函数形式做出了较强的假设。为了更好地理解这一点，

how these two variables are related, we visualize them together in Figure 4. We used generalized additive models (GAMs). GAMs are models that allow for non-linear relationships between predictor and response variables. We fit GAMs to predict reading times with smooth terms for IG, controlling for frequency, surprisal, contextual entropy, and (for English) word length.¹² We applied bootstrap smoothing over 20 resamples and computed confidence intervals for the estimated effects. We observe a consistent trend across both languages and all multimodal models: reading time remains relatively stable at lower IG estimates but decreases rapidly as IG increases in the upper end of its range. The Bayesian baseline does not show this pattern, as its IG values largely reflect lexical frequency.

如何这两个变量相关，我们在图 4 中将它们一起可视化。我们使用了广义可加模型（GAMs）。GAMs 是一种允许预测变量和响应变量之间存在非线性关系的模型。我们拟合了 GAMs 来预测阅读时间，对 IG 使用平滑项，并控制了频率、意外性、语境熵以及（对于英语）词长等因素。¹² 我们在 20 次重抽样上应用了自助平滑，并计算了估计效应的置信区间。我们观察到，在两种语言和所有多模态模型中都存在一致的趋势：阅读时间在较低的 IG 估计值下相对稳定，但在其范围的上端随着 IG 的增加而迅速下降。贝叶斯基线

模型没有表现出这种模式，因为其 IG 值主要反映了词汇频率。

5 Discussion

5 讨论

Turning back to our main prediction, we argue that our results provide converging evidence that visual quality, as measured by mutual information or information gain, impacts ease of processing. First, we find a consistent ordering, both in terms of reading times and mutual information, across our three experimental conditions. Second, we find a significant effect of the pointwise mutual information, or information gain, of individual words on reading times. While intuitive, the idea that bottom-up informational quality impacts ease of reading has not been quantified within a formal framework of reading. Our methods and experiments provide a specific estimate for the relationship between visual informational quality and reading times, which in English is between 25–66 ms/nat and in Chinese 5–10 ms/nat. However, these numbers should be taken only as rough estimates, as the exact functional form may not be linear.

回到我们主要的预测，我们认为我们的研究结果提供了趋同的证据，表明视觉质量（通过互信息或信息增益来衡量）会影响加工的容易程度。首先，我们在三种实验条件下都发现了一个一致的顺序，无论是从阅读时间还是从互信息来看。其次，我们发现单个词的点互信息或信息增益对阅读时间有显著影响。虽然这个想法很直观，但关于自下而上的信息质量影响阅读容易程度的观点尚未在阅读的正式框架中进行量化。我们的方法和实验为视觉信息质量与阅读时间之间的关系提供了一个具体的估计，其中在英语中，这一关系约为 25–66 ms/nat，在中文中约为 5–10 ms/nat。然而，这些数字只能作为粗略的估计，因为确切的功能形式可能不是线性的。

Turning now to our two sub research questions outlined in section 3.1: Interestingly, we find that information is not distributed evenly between the top and bottom halves of words. Both English and Chinese place more information about word identity in the top half of their orthographic systems, a feature which we argue is reflected in the quicker reading times for our Upper condition. This asymmetry may connect to more general visual biases, such as the top-heavy bias in object recognition (Viola Macchi et al., 2004) or the upper visual field

现在我们转向第 3.1 节中提出的两个子研究问题：有趣的是，我们发现信息在单词的上半部分和下半部分之间并不是均匀分布的。英语和汉语都在其拼写系统上半部分中包含更多关于单词身份的信息，这一特征我们认为体现在我们上半部分条件的阅读时间更快上。这种不对称性可能与更一般的视觉偏见有关，例如物体识别中的上重偏见（Viola Macchi 等，2004）或上视觉区域的偏好。

advantage (Previc, 1990), as well as the way writing systems are shaped by reading and writing direction.

优势 (Previc, 1990)，以及书写系统如何受到阅读和书写方向的影响。

Interestingly, Pimentel et al. (2021) find similar informational asymmetries between the beginnings and ends of words, using an even wider set of languages. Exploring whether their asymmetry in reading times and extending our results to more languages is an important direction for future research. Finally, we find some suggestive evidence that this asymmetry is stronger in English, reflected in the larger effect sizes for the Upper vs. Lower contrast in our reading data. Future work should investigate such differences in greater detail.

有趣的是，Pimentel 等人（2021）发现，在更广泛的语言集合中，单词的开头和结尾之间也存在类似的语义不对称性。探索他们的阅读时间不对称性，以及将我们的研究结果扩展到更多语言，是未来研究的重要方向。最后，我们发现了一些有说服力的证据表明，这种不对称性在英语中更为显著，这体现在我们阅读数据中“上部”与“下部”对比的效应量更大。未来的工作应更详细地研究此类差异。

Limitations

限制

There are several limitations with the present work. In our formal model, we made several assumptions: that visual samples of a given word E are drawn i.i.d. during reading, and that visual inputs are conditionally independent from each other given W . While these assumptions are strong, they are compatible with a “simple but fast” approach to reading. We discuss them in more detail in Section A. Moreover, our model assumes that the entropy threshold ϕ (Eq. 2) is always eventually reached, whereas in reality, readers may sometimes move on without reaching this threshold, especially under poor input quality. Besides, our model treats reading as a linear process, abstracting away from regressions, skips, and parafoveal preview. By focusing on FPRT as an index of early lexical processing, we capture word-level difficulty while necessarily ignoring the full dynamics of scanpaths. These simplifications keep the model straightforward but also mark directions for future work.

当前的工作存在一些限制。在我们的形式模型中，我们做了一些假设：在阅读过程中，给定单词 E 的视觉样本是独立同分布的 (i.i.d.)，并且在给定 W 的情况下，视觉输入之间是条件独立的。虽然这些假设较强，但它们与一种“简单但快速”的阅读方法是一致的。我们在附录 A 中更详细地讨论了这些假设。此外，我们的模型假设熵阈值 ϕ （公式 2）最终总是会被达到，但在现实中，读者有时可能在未达到这一阈值的情况下继续阅读，尤其是在输入质量较差的情况下。此外，我们的模型将阅读视为一个线性过程，忽略了回视、跳读和旁中央视野预览等现象。通过将 FPRT 作为早期词汇处理的指标，我们捕捉了单词层面的难度，但不可避免地忽略了扫视路径的完整动态。这些简化使模型更加直接，但也指明了未来工作的方向。

Another limitation concerns our approach to estimating mutual information between word identity and orthographic representation in Chinese. We used characters, rather than lexical words, as the unit of analysis. This choice was motivated by two considerations: first, the average word length in our OneStopQA Chinese dataset is approximately 1.4 characters; second, Chinese characters, unlike English letters, carry substantial visual and semantic complexity. As such, characters may serve as a more suitable unit for modeling bottom-up visual processing in Chinese, analogous to words in English. Nonetheless, using lexical words might produce slightly different estimates of mutual information. Future work could examine whether

另一个限制涉及我们对中文中词的身份与拼写表示之间互信息的估计方法。我们使用汉字而非词汇作为分析单位。这一选择基于两个考虑：首先，我们的 OneStopQA 中文数据集中，平均词长约为 1.4 个字符；其次，与英语字母不同，汉字在视觉和语义上具有显著的复杂性。因此，汉字可能更适合用于建模中文中的自下而上的视觉处理，类似于英语中的词汇。然而，使用词汇可能会产生略微不同的互信息估计值。未来的工作可以探讨是否

similar patterns hold when words are used instead of characters.

当使用词语代替字符时，类似的模式也适用。

Another limitation of the present work is that we did not extensively explore how top-down (contextual) processing can be integrated into the investigation of bottom-up processing. While much current research in psycholinguistics emphasizes the role of top-down expectations, our study is intended as a contribution specifically to the modeling of bottom-up processing. As described in Section 2.1, our formal Bayesian model is capable of incorporating contextual terms in a straightforward manner. However, for the scope of this analysis, we opted to exclude this component, leaving its exploration for future work.

另一个限制是，我们没有深入探讨如何将自上而下（上下文）处理整合到对自下而上处理的研究中。尽管当前心理语言学研究非常强调自上而下期望的作用，但我们的研究旨在特别为自下而上处理的建模做出贡献。如第 2.1 节所述，我们的形式化贝叶斯模型能够以一种直接的方式纳入上下文项。然而，在本次分析的范围内，我们选择排除了这一部分，将其探索留待未来的工作。

Moreover, our study is limited to English and Chinese, as they are chosen to represent alphabetic and logographic systems, given the available corpora and our expertise. While this provides

meaningful cross-linguistic insight, extending to more languages is important to generalize the conclusions of this work. In particular, the omission of Semitic languages is a notable gap, as their nonconcatenative morphology and distinct reading behaviors provide a critical test of generalizability (see e.g., Alhama et al., 2019 and Lerner et al., 2014). Including such languages remains an important direction for future work.

此外，我们的研究仅限于英语和汉语，因为它们被选为代表字母文字系统和意音文字系统的语言，这基于现有的语料库和我们的专业知识。虽然这提供了有意义的跨语言见解，但扩展到更多语言对于推广本研究的结论非常重要。特别是，闪含语系语言的缺失是一个显著的空白，因为它们的非连写形态和独特的阅读行为为一般性结论的可推广性提供了关键的检验（例如，参见 Alhama 等人，2019 年和 Lerner 等人，2014 年）。纳入此类语言仍然是未来研究的重要方向。

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Ethics Statement

伦理声明

We do not foresee any specific ethical concerns deriving from this work. Experimental participants engaged in standard reading tasks, and were compensated for their time. We used AI models for analysis. As with all AI systems, it is important to acknowledge a general risk of bias or misuse.

我们不预见到这项工作会带来任何特定的伦理问题。实验参与者参与了标准的阅读任务，并因其时间得到了补偿。我们使用了人工智能模型进行分析。如同所有人工智能系统一样，承认存在一般性的偏见或误用风险是很重要的。

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